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Interior Wall & Floor Segmentation

Whole Project in a glance

I have experimented with 4 approaches for this project, where two are from core computer vision and rest 2 are deep learning based.

The problem with only computer vision is that- most of the time it will manual as we are playing with thresholds. And for edge detection, gray scales of the images will reduce the info. While it is efficient and working pretty much good on the simpler problems, where the room is empty and not so much objects are there. When the walls and floor are separated by the straight line v1 works so well. As it detects through the horizontal edges who are greater than the threshold.

for v2 in core computer segment some non-linearity and more spatial intelligences added. But it performs poor. In simple cases it add un-wanted non-lineaty and in non-liner cases it is less capable.

Then it comes to the deep learning based approach, the v1 and v2 raw U-net segmenation models are trained on around 600 images from scratch. V1 encodes on 2d-cnn layers and v2 on resnet(weights = flase) with batch norm and early stopping.V1 performs well on simple sptial images and v2 works fine with more complex and noise conditions. However there are some signs of overfitting because of tiny dataset but patetions parmeter will stopped the training at 65 epochs. All the graphs and inference samples are below.

Project Structure

```
wall_floor_seg/
├── core_cv/
│   ├── v1/
│   └── v2/
├── deep_learning/
│   ├── v1/
│   └── v2/
├── dataset/
├── predictions/
└── README.md
```

1. Core Computer Vision based approches

The first stage was implemented without using a deep learning model. It has two versions - v1 and v2

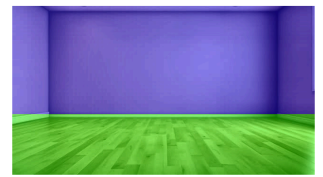
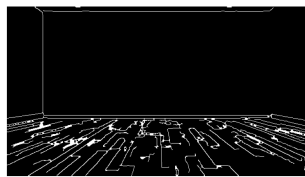
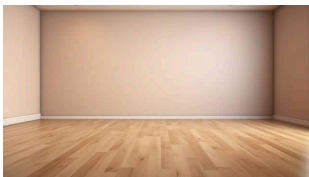
V1 — Geometric Approach

assumption :

wall and floor can generally be represented by a horizontal line which is large most of the cases.

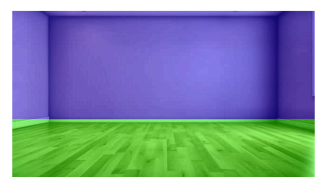
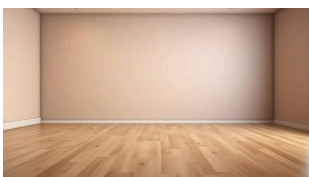
The pipeline was approximately:

```
Input Image
  ↓
Preprocessing
  ↓
Canny Edge Detection
  ↓
Morphology
  ↓
Lines Detection & hight estimation
  ↓
Horizontal Line Filtering
  ↓
Wall / Floor Boundary
  ↓
Segmentation
```



Limitation

It only creates liner boundries. Performe well on simple edge cases. But when it comes to a bit tougher non liner boundry. it failed.



V2 — Non-linearity + Spatiality

Improvements

- Color differences
- Pixel intensity variation
- Spatial regions
- Non-linear wall/floor boundaries

General pipeline:

```
Input Image
  ↓
Image Preprocessing
  ↓
Color / Intensity Analysis
  ↓
Region Segmentation
  ↓
Boundary Refinement
  ↓
Wall / Floor Mask
```

This approach produced significantly better segmentation compared with the first classical version, particularly in cases where the wall-floor boundary was not a clean horizontal line. But it also didn't seem perfect.

Limitations

However we are able to make the non-linear mask and some spatial understanding but that is still not much precise as the whole pipeline is based on the thresholding and manual interventions. so i decided to move toward deep-learning. While remembering that i'm not advised to use any pre-trained model or services. Here are a few examples of this v2 approach :



2. Deep Learning based approaches

After didn't getting the desired results. I Decded to move into the u-net model while keeping the constraints in mind. Training v1 and v2 u-net model on custome archetecture from use scratches with out any pre-defined weights. Both model will train only on 700 img dataset with collab T4 GPU. Here is the config of dataset-

```
Train: 626  
Val: 78  
Test: 79
```

V1 — U-Net with Custom CNN Encoder

The first deep learning version uses a standard U-Net style architecture with a custom CNN encoder.

Parameters

- Custom CNN encoder
- Loss - Cross Entropy Loss
- Optimizer - ADAM
- Convolution + Batch Normalization + ReLU blocks
- SEED = 42
- NUM_CLASSES = 3
- IMG_W, IMG_H = 512, 288
- BATCH_SIZE = 16
- EPOCHS = 100
- LR = 3e-4

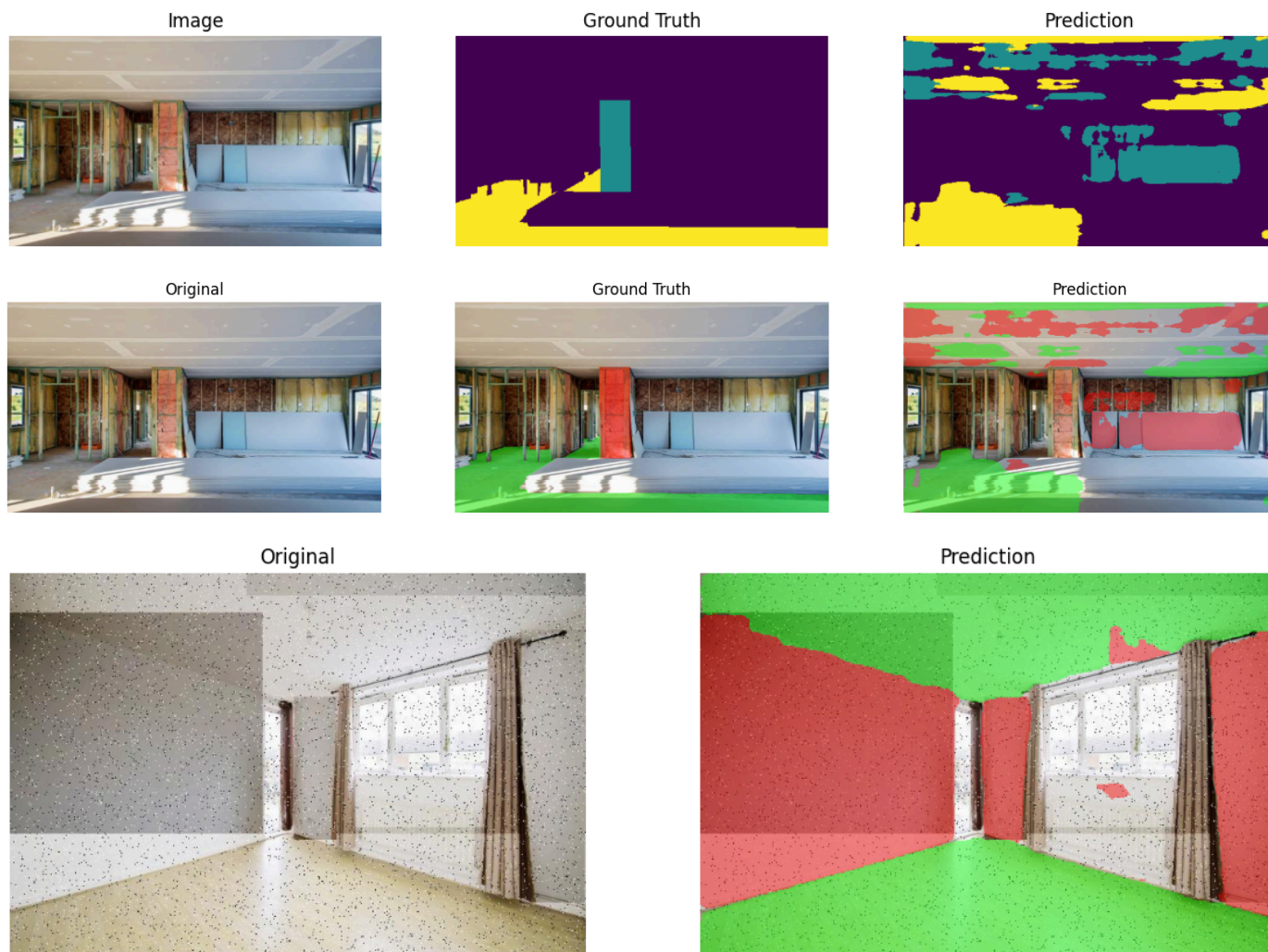
- WEIGHT_DECAY = $1e-4$
- PATIENCE = 12

Evals on Test Dataset

```
Test Loss: 0.4402932154981396
Wall IoU: 0.705119
Floor IoU: 0.6324628
Foreground mIoU: 0.66879094
Wall Dice: 0.82706136
Floor Dice: 0.7748572
```

Inference Samples

Majorly connected and big spatial structures readed well by model. Works well for the raw rooms where there are not many objects. In the noisy structures it struggles.



V2 — U-Net with ResNet Encoder

The custom CNN encoder is relatively simple and learns all visual features from the dataset. Using Resnet deeper feature extraction is possible as we trained from scratch so it required for quite better performance.

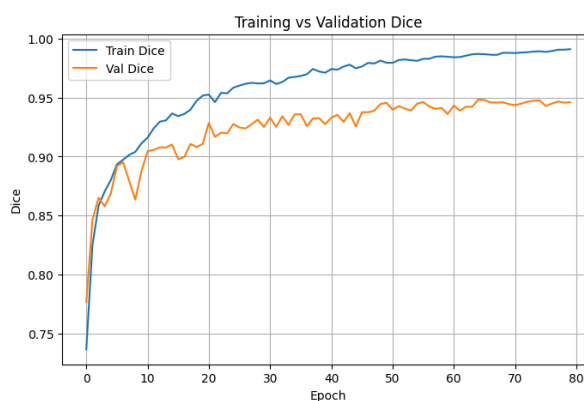
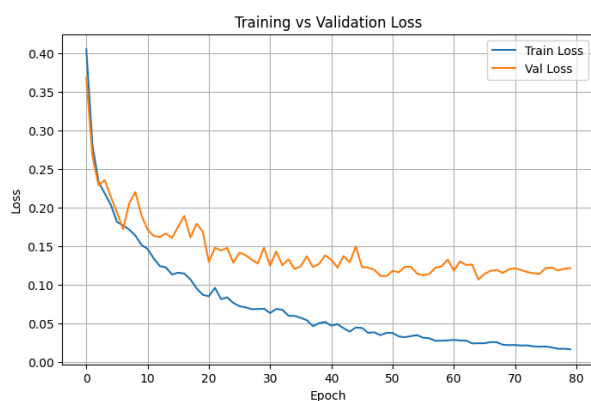
Architecture

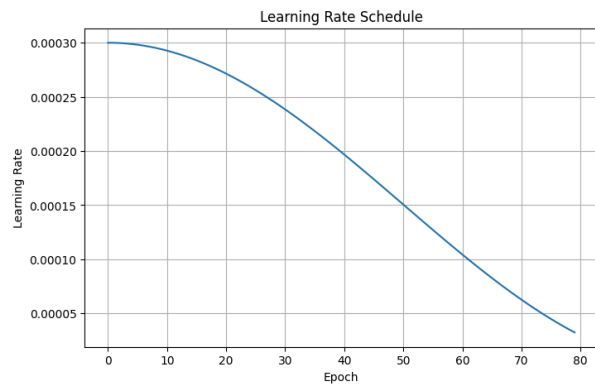
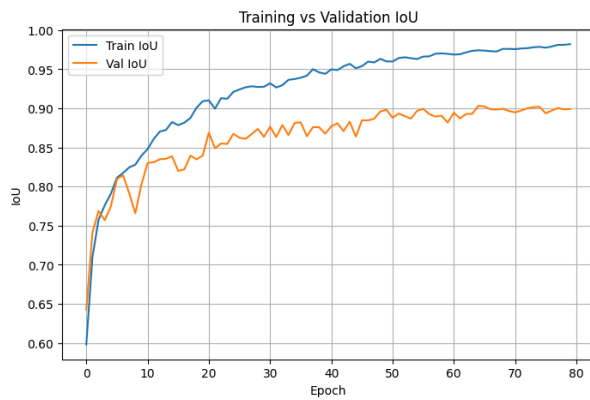
```
* Encoder - ResNet(weights = False)
* Loss - Cross Entropy Loss
* SEED = 42
* NUM_CLASSES = 2
* IMG_W = 512
* IMG_H = 288
* BATCH_SIZE = 16
* EPOCHS = 100
* LR = 3e-4
* WEIGHT_DECAY = 1e-4
* PATIENCE = 15
```

Eval on Test data

```
Test Loss : 0.1434
Test Dice : 0.9355
Test IoU : 0.8811
```

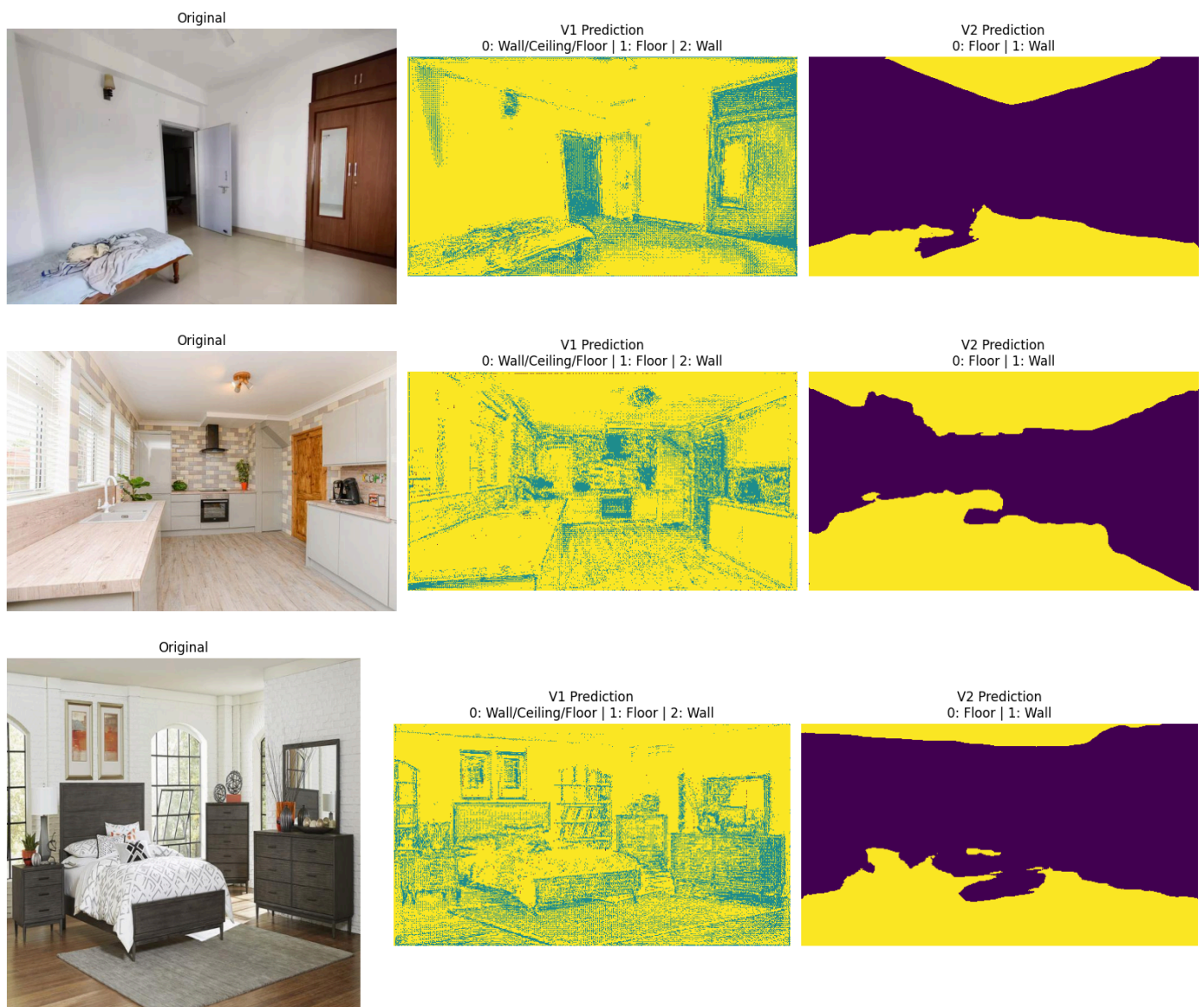
Training curves





some overfitting is there due to the very tiny data set but training will stopped on 65 epoches by early stopping param.

Infernces



Comparison in V1 and V2

Original



V1



V2



Original



V1



V2



Original



V1



V2

